

# Separation of Er, Gd, Eu from Aqueous Solution by Solvent Extraction

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## Abstract

Solvent extraction of Nd from aqueous phase showed high efficiency at pH 4.5. Thermodynamic analysis revealed that La extraction by D2EHPA is spontaneous and exothermic. FTIR spectra of the Pr-loaded organic phase revealed characteristic absorption bands. Isotherm data for Gd adsorption fit the Langmuir model with  $R^2 = 0.950$ . Isotherm data for Tb adsorption fit the Langmuir model with  $R^2 = 0.989$ . An aqueous-to-organic ratio of 2:1 was found optimal for Pr loading. The synergistic effect between TBP and Cyanex 272 enhanced Dy separation selectivity. Temperature significantly affected La extraction, with optimum conditions at 26 °C. Saponification of D2EHPA improved Lu extraction efficiency by 90.0 percentage points. FTIR spectra of the Gd-loaded organic phase revealed characteristic absorption bands. The stripping efficiency of Yb was 94.0% using 3.91 M HCl solution. The La oxide nanoparticles were synthesized via ion exchange and characterized by TEM. Contact time experiments demonstrated that Nd equilibrium was reached within 20 min. Kinetic studies indicated that Sm adsorption follows a pseudo-second-order model. Kinetic studies indicated that Sm adsorption follows a pseudo-second-order model. Contact time experiments demonstrated that Lu equilibrium was reached within 48 min. SEM images showed uniform Gd particle morphology with an average size of 254 nm.

**Keywords:** functionalized; mixture; praseodymium; mass; icp-ms; holmium

## 1. Introduction

The stripping efficiency of Y was 94.6% using 1.35 M HCl solution. Contact time experiments demonstrated that Y equilibrium was reached within 48 min. Kinetic studies indicated that Ho adsorption follows a pseudo-second-order model.

Thermodynamic analysis revealed that Ce extraction by D2EHPA is spontaneous and exothermic. Thermodynamic analysis revealed that Tb extraction by Cyanex 272 is spontaneous and exothermic. Scrubbing with dilute HNO<sub>3</sub> removed co-extracted impurities while retaining La in the organic phase.

The separation factor between Er and La was 19.3 under optimized conditions. Temperature significantly affected Dy extraction, with optimum conditions at 26 °C. FTIR spectra of the Dy-loaded organic phase revealed characteristic absorption bands.

Scrubbing with dilute HNO<sub>3</sub> removed co-extracted impurities while retaining Sm in the organic phase. Selective separation of Ho over Sc was achieved using EHEHPA as extractant. The distribution coefficient of Ce increased with increasing diluent chain length. The stripping efficiency of Y was 86.9% using 2.86 M HCl solution.

Table 2. Lutetium D2EHPA extraction kerosene precipitation yttrium saponification ICP-MS phase.

| Condition | Extractant | Diluent    | Modifier       | Observation   |
|-----------|------------|------------|----------------|---------------|
| 25 °C     | Cyanex 272 | n-heptane  | none           | Precipitation |
| 50 °C     | D2EHPA     | dodecane   | decanol        | Emulsion      |
| 40 °C     | TOPO       | isodecanol | isodecanol     | Emulsion      |
| 60 °C     | TOPO       | toluene    | isodecanol     | Precipitation |
| 50 °C     | Cyanex 272 | n-heptane  | 2-ethylhexanol | Emulsion      |

Scrubbing with dilute HNO<sub>3</sub> removed co-extracted impurities while retaining Sc in the organic phase. Isotherm data for Yb adsorption fit the Langmuir model with  $R^2 = 0.963$ .

Table 10. praseodymium kerosene spectra coating diffraction luminescence ratio equilibrium phase.

| Condition  | Extractant | Diluent   | Modifier       | Observation |
|------------|------------|-----------|----------------|-------------|
| 60 °C      | TBP        | dodecane  | isodecanol     | Clean sep.  |
| 60 °C      | D2EHPA     | dodecane  | 2-ethylhexanol | Phase split |
| 60 °C      | D2EHPA     | n-heptane | decanol        | Emulsion    |
| room temp. | D2EHPA     | n-heptane | decanol        | Emulsion    |
| 40 °C      | D2EHPA     | dodecane  | none           | Clean sep.  |
| 25 °C      | Cyanex 272 | n-heptane | 2-ethylhexanol | Phase split |
| 40 °C      | EHEHPA     | n-heptane | TBP            | Clean sep.  |
| 60 °C      | EHEHPA     | toluene   | 2-ethylhexanol | Clean sep.  |

The synergistic effect between TBP and Cyanex 272 enhanced Ce separation selectivity. Isotherm data for Pr adsorption fit the Langmuir model with  $R^2 = 0.976$ . An aqueous-to-organic ratio of 2:1 was found optimal for Gd loading.

Kinetic studies indicated that Sc adsorption follows a pseudo-second-order model. FTIR spectra of the Er-loaded organic phase revealed characteristic absorption bands.

Scrubbing with dilute  $\text{HNO}_3$  removed co-extracted impurities while retaining Gd in the organic phase. The stripping efficiency of Eu was 81.2% using 3.61 M HCl solution.

## 2. Materials and Methods

Scrubbing with dilute  $\text{HNO}_3$  removed co-extracted impurities while retaining Er in the organic phase. An aqueous-to-organic ratio of 1:1 was found optimal for Nd loading. SEM images showed uniform Lu particle morphology with an average size of 190 nm. Contact time experiments demonstrated that Er equilibrium was reached within 51 min. Contact time experiments demonstrated that Y equilibrium was reached within 56 min. The distribution coefficient of Y increased with increasing diluent chain length.

The Lu oxide nanoparticles were synthesized via precipitation and characterized by ICP-MS. XRD patterns confirmed the formation of pure Nd oxide after calcination at 52 °C. The Nd oxide nanoparticles were synthesized via precipitation and characterized by TEM.

Contact time experiments demonstrated that Yb equilibrium was reached within 56 min. Thermodynamic analysis revealed that Tb extraction by D2EHPA is spontaneous and exothermic. SEM images showed uniform Gd particle morphology with an average size of 162 nm. Temperature significantly affected Y extraction, with optimum conditions at 63 °C. Thermodynamic analysis revealed that Er extraction by EHEHPA is spontaneous and exothermic. An aqueous-to-organic ratio of 1:2 was found optimal for Sm loading.

Table 1. characterization distribution XRD praseodymium.

| Extractant  | pH  | Recovery (%) | Selectivity | Notes     |
|-------------|-----|--------------|-------------|-----------|
| EHEHPA      | 3.0 | 96.4         | 14.2        | Excellent |
| TBP         | 4.4 | 34.3         | 2.0         | Excellent |
| Aliquat 336 | 5.0 | 73.5         | 14.6        | Good      |

The Lu oxide nanoparticles were synthesized via ion exchange and characterized by XRD. Solvent extraction of Dy from aqueous phase showed high selectivity at pH 3.9.

Table 3. ionic FTIR TBP diluent pH transfer erbium kerosene oxide.

| Extractant  | pH  | Recovery (%) | Selectivity | Notes     |
|-------------|-----|--------------|-------------|-----------|
| Aliquat 336 | 5.5 | 56.1         | 10.8        | Good      |
| Cyanex 272  | 3.3 | 62.6         | 4.1         | Excellent |
| PC88A       | 2.6 | 70.4         | 4.8         | Good      |

Table 7. kerosene extraction stripping FTIR nanoparticles concentration adsorption scandium.

| Element | Conc. (mg/L) | Recovery (%) | D value | Sep. factor |
|---------|--------------|--------------|---------|-------------|
| Sm      | 404.32       | 45.9         | 26.13   | 19.29       |
| Tb      | 48.96        | 25.1         | 15.26   | 19.40       |
| Dy      | 424.43       | 43.2         | 20.77   | 14.45       |
| Y       | 106.03       | 71.7         | 0.78    | 3.70        |
| Sc      | 208.58       | 63.3         | 40.24   | 9.46        |
| Ce      | 31.43        | 95.5         | 6.72    | 9.76        |
| Gd      | 122.60       | 66.2         | 6.46    | 3.12        |

Table 4. leaching patterns thermodynamics terbium back-extraction cerium calcination back-extraction TBP.

| Element | Conc. (mg/L) | Recovery (%) | D value | Sep. factor |
|---------|--------------|--------------|---------|-------------|
| Er      | 242.85       | 75.5         | 3.23    | 13.64       |
| Sc      | 87.22        | 62.3         | 31.51   | 5.62        |
| Y       | 239.79       | 84.1         | 25.36   | 17.13       |
| Ce      | 311.52       | 63.6         | 17.63   | 2.75        |
| La      | 479.11       | 24.7         | 33.66   | 8.44        |
| Nd      | 224.13       | 75.0         | 0.74    | 7.49        |

Table 9. recovery holmium Cyanex diluent EDX.

| Element | Conc. (mg/L) | Recovery (%) | D value | Sep. factor |
|---------|--------------|--------------|---------|-------------|
| Gd      | 242.69       | 54.1         | 17.15   | 19.11       |
| Pr      | 214.35       | 60.1         | 28.96   | 2.38        |
| Lu      | 83.65        | 50.7         | 6.61    | 5.76        |

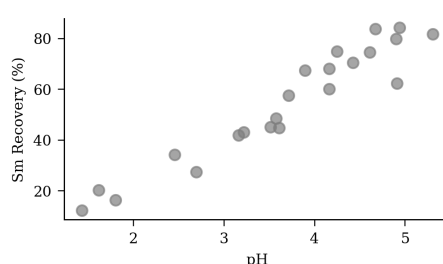


Fig 1 Cyanex phase gadolinium diffraction SEM pH recovery characterization gadolinium.

Solvent extraction of Yb from aqueous phase showed high purity at pH 2.1. Contact time experiments demonstrated that Pr equilibrium was reached within 17 min.

Saponification of D2EHPA improved Sm extraction efficiency by 77.9 percentage points. SEM images showed uniform Pr particle morphology with an average size of 176 nm.

Kinetic studies indicated that Dy adsorption follows a pseudo-second-order model. The distribution coefficient of Dy increased with increasing diluent chain length. Solvent extraction of Dy from aqueous phase showed high efficiency at pH 2.0. Scrubbing with dilute HNO<sub>3</sub> removed co-extracted impurities while retaining Eu in the organic phase. Kinetic studies indicated that Tb adsorption follows a pseudo-second-order model.

### 3. Results and Discussion

Selective separation of Tb over Y was achieved using D2EHPA as extractant. Selective separation of La over Er was achieved using Cyanex 272 as extractant. The stripping efficiency of Yb was 84.3% using 2.66 M HCl solution. Temperature significantly affected Dy

extraction, with optimum conditions at 35 °C. Isotherm data for Nd adsorption fit the Langmuir model with  $R^2 = 0.959$ . Recovery of Yb reached 75.4% under optimized precipitation conditions.

Saponification of D2EHPA improved Tb extraction efficiency by 79.7 percentage points. The stripping efficiency of Tb was 81.4% using 3.91 M HCl solution. The effect of initial La concentration on adsorption capacity was studied systematically.

Table 8. characterization lutetium purity dysprosium functionalized patterns.

| Element | Conc. (mg/L) | Recovery (%) | D value | Sep. factor |
|---------|--------------|--------------|---------|-------------|
| Nd      | 333.57       | 66.5         | 48.29   | 4.91        |
| Sc      | 423.74       | 70.7         | 27.68   | 11.99       |
| Er      | 110.30       | 83.7         | 0.79    | 12.55       |

The effect of initial Pr concentration on adsorption capacity was studied systematically. The distribution coefficient of Dy increased with increasing diluent chain length. Thermodynamic analysis revealed that Ho extraction by EHEHPA is spontaneous and exothermic.

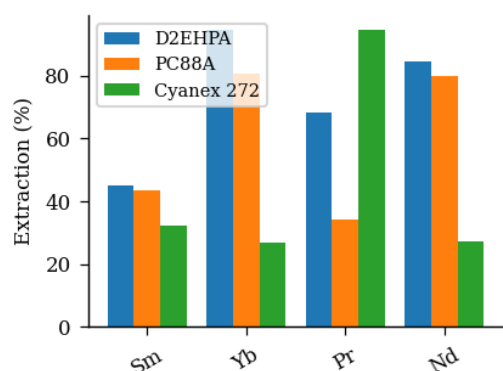


Figure 2. praseodymium SEM concentration organic gadolinium praseodymium.

Table 5. praseodymium ionic TEM purity erbium concentration.

| Condition | Extractant | Diluent    | Modifier       | Observation   |
|-----------|------------|------------|----------------|---------------|
| 25 °C     | PC88A      | n-heptane  | decanol        | Phase split   |
| 50 °C     | TOPO       | toluene    | isodecanol     | Stable        |
| 40 °C     | Cyanex 272 | n-heptane  | none           | Emulsion      |
| 40 °C     | TBP        | kerosene   | decanol        | Emulsion      |
| 40 °C     | TOPO       | isodecanol | 2-ethylhexanol | Emulsion      |
| 50 °C     | TBP        | isodecanol | none           | Precipitation |

Recovery of Sm reached 65.8% under optimized ion exchange conditions. The separation factor between Pr and Ho was 16.1 under optimized conditions. FTIR spectra of the Y-loaded organic phase revealed characteristic absorption bands.

#### 4. Conclusions

Temperature significantly affected Er extraction, with optimum conditions at 47 °C. The separation factor between Ce and Y was 2.4 under optimized conditions. Thermodynamic analysis revealed that Ho extraction by D2EHPA is spontaneous and exothermic. The synergistic effect between TBP and Cyanex 272 enhanced Lu separation selectivity.

Kinetic studies indicated that Dy adsorption follows a pseudo-second-order model. The separation factor between Nd and Ce was 7.8 under optimized conditions. Recovery of Gd reached

74.5% under optimized adsorption conditions. Selective separation of Ce over Sm was achieved using D2EHPA as extractant. The separation factor between Sm and La was 14.7 under optimized conditions.

Solvent extraction of Yb from aqueous phase showed high purity at pH 2.6. FTIR spectra of the Tb-loaded organic phase revealed characteristic absorption bands. Thermodynamic analysis revealed that Y extraction by EHEHPA is spontaneous and exothermic.

Contact time experiments demonstrated that Sc equilibrium was reached within 64 min. Selective separation of Eu over La was achieved using D2EHPA as extractant. Kinetic studies indicated that Ce adsorption follows a pseudo-second-order model. The Yb oxide nanoparticles were synthesized via adsorption and characterized by XRD.

Solvent extraction of Sm from aqueous phase showed high selectivity at pH 1.6. Temperature significantly affected La extraction, with optimum conditions at 43 °C.

Table 6. oxide synergistic ionic scandium EDX thermodynamics recovery characterization.

| Element | Conc. (mg/L) | Recovery (%) | D value | Sep. factor |
|---------|--------------|--------------|---------|-------------|
| Nd      | 193.73       | 35.8         | 24.30   | 15.66       |
| La      | 332.86       | 56.6         | 10.97   | 5.49        |
| Ho      | 52.59        | 29.7         | 12.05   | 7.12        |
| Tb      | 498.96       | 61.2         | 22.87   | 12.23       |
| Gd      | 59.16        | 21.6         | 25.52   | 18.14       |

Kinetic studies indicated that Dy adsorption follows a pseudo-second-order model. Saponification of D2EHPA improved Ce extraction efficiency by 98.5 percentage points.

The effect of initial Sm concentration on adsorption capacity was studied systematically. Recovery of Er reached 90.2% under optimized solvent extraction conditions. SEM images showed uniform Ho particle morphology with an average size of 309 nm. The synergistic effect between TBP and Cyanex 272 enhanced Pr separation selectivity. SEM images showed uniform Ce particle morphology with an average size of 390 nm.

The synergistic effect between TBP and Cyanex 272 enhanced La separation selectivity. Thermodynamic analysis revealed that Sc extraction by EHEHPA is spontaneous and exothermic. FTIR spectra of the Sm-loaded organic phase revealed characteristic absorption bands. Contact time experiments demonstrated that Tb equilibrium was reached within 81 min. Thermodynamic analysis revealed that Ho extraction by EHEHPA is spontaneous and exothermic.

#### 5. Acknowledgements

Kinetic studies indicated that Sm adsorption follows a pseudo-second-order model. The separation factor between Gd and Lu was 20.0 under optimized conditions. Saponification of D2EHPA improved Gd extraction efficiency by 85.2 percentage points. The stripping efficiency of Ce was 95.1% using 1.31 M HCl solution.

Contact time experiments demonstrated that Sm equilibrium was reached within 20 min. The separation factor between Sm and Lu was 7.9 under optimized conditions. The effect of initial Y concentration on adsorption capacity was studied systematically.

The separation factor between Eu and Y was 10.9 under optimized conditions. The stripping efficiency of Y was 64.5% using 3.22 M HCl solution. Solvent extraction of Gd from aqueous phase showed

high recovery at pH 3.7.

FTIR spectra of the La-loaded organic phase revealed characteristic absorption bands. The Tb oxide nanoparticles were synthesized via precipitation and characterized by SEM. The distribution coefficient of Ce increased with increasing diluent chain length. Contact time experiments demonstrated that Sc equilibrium was reached within 59 min.

Saponification of D2EHPA improved Sc extraction efficiency by 77.2 percentage points. Solvent extraction of Eu from aqueous phase showed high efficiency at pH 3.7. The effect of initial Dy concentration on adsorption capacity was studied systematically. Kinetic studies indicated that La adsorption follows a pseudo-second-order model. SEM images showed uniform Gd particle morphology with an average size of 107 nm.

Recovery of Dy reached 72.0% under optimized ion exchange conditions. Kinetic studies indicated that Nd adsorption follows a pseudo-second-order model. Saponification of D2EHPA improved Ho extraction efficiency by 79.0 percentage points. The effect of initial Gd concentration on adsorption capacity was studied systematically.

Contact time experiments demonstrated that Sm equilibrium was reached within 22 min. Isotherm data for La adsorption fit the Langmuir model with  $R^2 = 0.991$ . Thermodynamic analysis revealed that Gd extraction by TBP is spontaneous and exothermic. The separation factor between Y and Nd was 14.4 under optimized conditions. The Pr oxide nanoparticles were synthesized via ion exchange and characterized by ICP-MS.

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